

Ahmed Nesar Tahsin Choudhury

tahsinbd247@gmail.com | tahsinchoudhury.github.io | [LinkedIn](#)

[GitHub](#) | [Codeforces](#) | [Google Scholar](#)

Education

University of Dhaka, BSc in Computer Science and Engineering 2022 – 2026

- **CGPA:** 3.97/4.00
- **Rank:** 1 out of 65
- **Admission Test:** 59th out of ~90,000 applicants (**99.9th** Percentile)
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Mathematical and Statistical Analysis for Engineers, Introduction to Probability and Statistics, Parallel and Distributed Systems, Introduction to Numerical Methods, Design and Analysis of Algorithms, Data Structures and Algorithms, Graph Theory, Introduction to Software Engineering

Research Experience

Research Assistant, Cognitive Agents and Interaction Lab, University of Dhaka June 2024 – Present

- Developed a state-of-the-art Bengali Handwritten Text Recognition System
- Experimented with encoder-decoder and decoder-only architectures in OCR
- Experimented with grapheme-level tokenization as an alternative to byte-level tokenization for complex scripts
- Showed that integrating linguistic context into tokenizers improves text recognition
- Currently working on training-free Video Anomaly Detection (VAD) using an event-level approach

Publications

Note: * indicates equal contribution

[GraDeT-HTR: A Bengali Handwritten Text Recognition System utilizing Grapheme-based Tokenizer and Decoder-only Transformer](#) Sept 2025

Md. Mahmudul Hasan*, **Ahmed Nesar Tahsin Choudhury***, Mahmudul Hasan, Md. Mosaddek Khan

Published at 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations

Deployment: <https://cognistorm.ai/hcr>

Professional Experience

Associate Machine Learning Engineer, [Therap Services](#) March 2026 – Present

- Currently conducting a systematic literature review on PII data masking as part of an ongoing company project
- Completed onboard training covering end-to-end development and optimization of modern language models
- Implemented a mini LLaMA-style language model from scratch and ran systematic evaluations
- Built retrieval-augmented generation and summarization pipelines for downstream NLP tasks
- Performed LoRA fine-tuning, RLHF with Direct Preference Optimization, and model quantization

Software Engineer Intern, [Mass Holdings Inc.](#) – Tokyo, Japan (Remote) July 2024 – Dec 2025

- Built a sales forecasting model with XGBoost to predict demand trends and improve decision-making
- Developed a small-scale live streaming prototype utilizing Nginx as the media server to process video streams through RTMP, delivering HLS segments
- Engineered a ticketing system featuring client ID generation and QR code verification for enhanced event management
- Designed a real-time interactive 3D object visualizer using smartphone cameras and Three.js, enabling clients to preview product fit within intended environments before purchasing

- Built foundational features for an e-commerce platform's backend, including carts, promotions, and order management, leveraging Spring Boot and MySQL

Awards and Achievements

- **2024: Runner-up**, CODE Samurai (National Hackathon)
- **2023:**
 - **26th** out of 196 teams in **NCPC Onsite 2023**, National Collegiate Programming Contest, JU
 - **24th** out of 96 teams in **CUET IUPC 2023**, Inter University Programming Contest 2023, CUET
 - **Champion** among 100 participants, Battle of Brains, University of Dhaka
- **2022:**
 - **Champion** (among freshmen), Battle of Brains, University of Dhaka
 - **45th** out of **1700+** teams, ICPC Dhaka Regionals Preliminary Round
- **2021: 59th** out of **~90,000 (99.9th** percentile) applicants, Admission Test – A unit, University of Dhaka
- **2020: General Board Scholarship Recipient**, HSC
- **2020: 8th** out of 393 participants in NHSPC, **National Camper**, National High School Programming Contest
- **2019: 11th** out of 1000+ participants in BDPhO, **National Camper**, Bangladesh Physics Olympiad
- **2018: Talentpool Board Scholarship Recipient**, SSC
- **2015: Talentpool Board Scholarship Recipient**, JSC

Skills

Competitive Programming: Codeforces Max Rating: 1588 (Username: Philoomath)

ML/DL: NumPy, Pandas, PyTorch

Coding: Java, Python, C, C++, JavaScript, SQL, $\LaTeX$, ...

Databases: MySQL, PostgreSQL

Web Development: Spring Boot, FastAPI, HTML, CSS, React JS (basics), ...

Miscellaneous: Tutoring high-school and college students

Languages: Bangla, English